

Experiment#01

Objective:

To study the introduction of lab equipment used in Digital Systems.

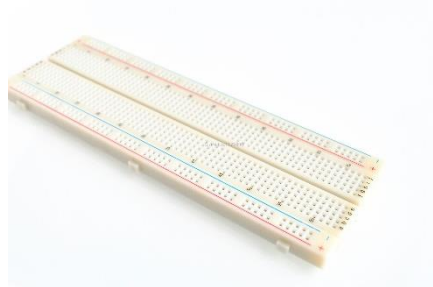
Equipments:

- Breadboard
- Jumper wire
- Integrated (IC's)
- Dc power supply
- Led ligh

Theory:

Breadboard:

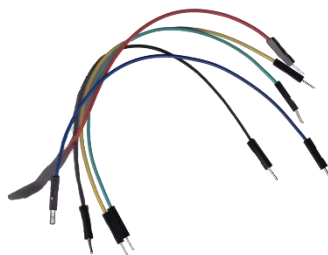
A breadboard is a construction base used to build semi-permanent prototypes of electronic circuits. These boards have rows of holes that can hold electronic components and connect them with wires without soldering. This allows for easy experimentation and modification of circuits, making breadboards popular for learning electronics, hobbyists, and prototyping.



Fig#01

Jumper-wires:

A jumper wire is an electrical wire with a connector or pin at each end. These wires are used to interconnect components on a breadboard or other prototype circuits without the need for soldering. Jumper wires allow you to easily connect different points in a circuit. They are commonly used with breadboards and other prototyping tools, making it simple to modify a circuit as needed.



Fig#02

DC power supply:

A DC power supply, is a device that produces a regulated direct current (DC) voltage output for electronic and electric devices. Unlike alternating current (AC) voltage, which can be stepped up or down using a transformer, DC voltage remains constant.



Fig#03

Integrated (IC's):

An integrated circuit (IC), also known as a microchip or chip, is a tiny electronic device that combines hundreds to billions of miniaturized electronic components, like transistors, resistors, and capacitors, onto a single piece of semiconductor material, usually silicon. These components are interconnected within the silicon chip using a complex series of layers and pathways, forming complete circuits without the need for separate wires or bulky components.

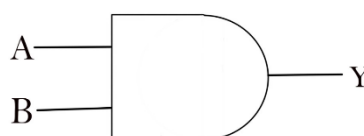
And-gate:

An AND gate is a fundamental logic gate in digital electronics that performs the logical AND operation on its inputs. In simpler terms, it acts like a gatekeeper, only allowing the output to pass through if all of its inputs are true (represented by the value 1). If even one input is false (represented by the value 0), the output remains blocked (at 0).

Expression:

$$Y = A \cdot B$$

Symbol:



Fig#04

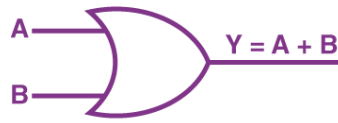
OR -gate:

An OR gate is a fundamental logic gate that performs the logical OR operation. In simple terms, it outputs a HIGH (true) signal if at least one of its inputs is HIGH, and a LOW (false) signal only if both inputs are LOW.

Expression:

$$Y = A + B$$

Symbol:



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Fig#05

Not-gate:

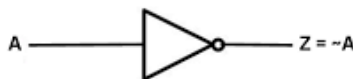
A NOT gate, sometimes called an inverter, is a fundamental building block in digital logic circuits. It's a simple yet powerful logic gate with just one input and one output. Its sole purpose is to flip the input signal, meaning:

- If the input is HIGH (1), the output becomes LOW (0).
- If the input is LOW (0), the output becomes HIGH (1).

Expression:

$$Y = \neg A$$

Symbol:



Fig#06

Conclusion:

In this experiment we know about the breadboard, jumper wires, DC power supply and integrated (IC's) and also about the different types of gates, its truth table and expression.